⑮ $\sum \frac{\cos n}{n^{2/3}+(-1)^n}$

$n^{2/3}+(-1)^n \sim n^{2/3}$, но не можем да заместим $n^{2/3}+(-1)^n$ с $n^{2/3}$, понеже $\cos n$ си мени знака. Но все пак знаем, че $\sum \frac{\cos n}{n^{2/3}}$ е сходящ и това е главната част на дадения ред. Затова я отделяме.

$$\frac{\cos n}{n^{2/3}+(-1)^n} = \frac{\cos n}{n^{2/3}} \frac{1}{1+\frac{(-1)^n}{n^{2/3}}} = \frac{\cos n}{n^{2/3}}\left(1-\frac{(-1)^n}{n^{2/3}}+o\left(\frac{1}{n^{2/3}}\right)\right) = \frac{\cos n}{n^{2/3}} + \frac{\cos n}{n^{2/3}} O\left(\frac{1}{n^{2/3}}\right)$$

Но $\left|\frac{\cos n}{n^{2/3}} O\left(\frac{1}{n^{2/3}}\right)\right| = O\left(\frac{1}{n^{4/3}}\right)$ и $\sum \frac{1}{n^{4/3}}$ е сх. $\sum \frac{\cos n}{n^{2/3}}$ също е сходящ.

Сравнете с $\sum \frac{\cos n}{n^\alpha}$ – сходящ при $\alpha>0$.

! Критериите на Лайбниц, Абел и Дирихле са неприложими, понеже няма монотонност.

⑯ $\sum (\sqrt[n]{n}-1)\cos n$

$$\sqrt[n]{n}-1 = n^{1/n}-1 = e^{\frac{1}{n}\ln n}-1 = \left(1+\frac{\ln n}{n}+O\left(\frac{\ln^2 n}{n^2}\right)\right)-1 = \frac{\ln n}{n}+O\left(\frac{\ln^2 n}{n^2}\right)$$

$$\Rightarrow \sum = \sum \frac{\ln n}{n}\cos n + \sum O\left(\frac{\ln^2 n}{n^2}\right)\cos n$$

$\sum \left|O\left(\frac{\ln^2 n}{n^2}\right)\cos n\right| = O\left(\sum \frac{\ln^2 n}{n^2}\right)$ – сходящ (дори абсолютно)

$f(x) = \frac{\ln x}{x} \Rightarrow f' = \frac{1-\ln x}{x^2}$ и при $x \geq 3 \Rightarrow f'(x)<0 \Rightarrow f \searrow$,

т.е. $\frac{\ln n}{n} \searrow 0$. Също $\left|\sum_1^n \cos n\right| < M$ – равномерно ограничени

$\Rightarrow$ критерият на Дирихле е приложим $\Rightarrow \sum$ е сх.

⑰ $\sum \left(e^{\cos n/\sqrt{n}} - \cos\frac{1}{n}\right)$

$$\frac{\cos n}{\sqrt{n}} \xrightarrow[n\to\infty]{} 0 \Rightarrow e^{\cos n/\sqrt{n}} = 1+\frac{\cos n}{\sqrt{n}}+\frac{1}{2}\frac{\cos^2 n}{n}+O\left(\frac{1}{n^{3/2}}\right)$$

$$\cos\frac{1}{n} = 1-\frac{1}{2n^2}+o\left(\frac{1}{n^2}\right) = 1+O(1/n^2)$$

$$\Rightarrow e^{\cos n/\sqrt{n}} - \cos\frac{1}{n} = \frac{\cos n}{\sqrt{n}}+\frac{1}{2}\frac{\cos^2 n}{n}+O\left(\frac{1}{n^{3/2}}\right)$$

$\sum \frac{\cos n}{\sqrt{n}}$ – сх. Сравнете с $\sum \frac{\cos n}{n^\alpha} \rightarrow$ сх. при $\alpha>0$

$\sum O\left(\frac{1}{n^{3/2}}\right) = O\left(\sum \frac{1}{n^{3/2}}\right)$ – сх. (дори абсолютно)

$$\sum \frac{\cos^2 n}{2n} = \sum \frac{1+\cos(2n)}{4n} = \frac{1}{4}\underbrace{\sum \frac{1}{n}}_{\text{разх.}} + \underbrace{\sum \frac{\cos(2n)}{4n}}_{\text{сх. сравнете с } \sum \frac{\cos n}{n^\alpha}}$$

Окончателно $\sum$ е разходящ.

⑱ $\sum \frac{1}{\sqrt[4]{n}} \cos\frac{\pi(n^2+1)}{n}$

$$\cos\frac{\pi(n^2+1)}{n} = \cos\left(\pi n+\frac{\pi}{n}\right) = (-1)^n\cos\frac{\pi}{n} = (-1)^n\left(1-\frac{\pi^2}{2n^2}+o\left(\frac{1}{n^2}\right)\right) = (-1)^n\left(1+O\left(\frac{1}{n^2}\right)\right)$$

$$\Rightarrow \sum = \sum \frac{(-1)^n}{\sqrt[4]{n}} + \sum \frac{(-1)^n}{\sqrt[4]{n}} O\left(\frac{1}{n^2}\right)$$

$\sum \frac{(-1)^3}{\sqrt[4]{n}}$ – сх. на Лайбниц: $\sum (-1)^n a_n$, $a_n \searrow 0 \Rightarrow$ сх.

$\sum \left|\frac{(-1)^3}{\sqrt[4]{n}} O\left(\frac{1}{n^2}\right)\right| = O\left(\sum \frac{1}{n^{9/4}}\right)$ – сх. (дори абсолютно)

⑲ $$\sum \frac{\sin(n+1/n)}{\ln\ln(n+1)} = \sum \frac{\sin n\cos\frac{1}{n}+\cos n\sin\frac{1}{n}}{\ln\ln(n+1)} = \sum_1 \frac{\sin n\cos\frac{1}{n}}{\ln\ln(n+1)} + \sum_2 \frac{\cos n\sin\frac{1}{n}}{\ln\ln(n+1)}$$

$$\sum_1 \rightarrow \cos\frac{1}{n} = 1+O\left(\frac{1}{n^2}\right) \Rightarrow \sum_1 = \sum_{11}\frac{\sin n}{\ln\ln(n+1)} + \sum_{12}\frac{\sin n}{\ln\ln(n+1)} O(1/n^2)$$

$\sum_{12} = O\left(\sum \frac{1}{n^2}\right) \rightarrow$ абс. сх., $\sum_{11}$ – сх. Следва от критерия на Дирихле понеже $\left|\sum_1^N \sin n\right| \leq M$, т.е. равномерно ограничени и $\frac{1}{\ln\ln(n+1)} \searrow 0$

$$\sum_2 = \sum \frac{\cos n}{\ln\ln(n+1)}\left(\frac{1}{n}+O\left(\frac{1}{n^3}\right)\right) = \sum_{21}\frac{\cos n}{\ln\ln(n+1).n} + \sum_{22}\frac{\cos n}{\ln\ln(n+1)} O\left(\frac{1}{n^3}\right)$$

$\sum_{21}$ – сх. отново понеже $\left|\sum_1^N \cos n\right| \leq M$ и $\frac{1}{n\ln\ln(n+1)} \searrow 0$, т.е. Дирихле

$\sum_{22} = O\left(\frac{1}{n^3}\right)$ – абс. сх. Окончателно $\sum$ – сходящ

⑳ $\sum (-1)^n \frac{\sqrt[3]{n+1}}{\sqrt{n}+2}$

Тъй като $\frac{\sqrt[3]{n+1}}{\sqrt{n}+2} \sim \frac{\sqrt[3]{n}}{\sqrt{n}} = \frac{1}{n^{1/6}}$ и $\sum \frac{(-1)^n}{n^6}$ е сх. ($\Leftarrow$ Лайбниц), то отделяме този ред.

! Ако $\sum(a_n+b_n) = \sum a_n + \sum b_n$ и $\sum b_n$ е сходящ, то $\sum a_n$ и $\sum(a_n+b_n)$ са едновременно сходящи или разходящи.

! Не можем директно да заместим в началния ред $\frac{\sqrt[3]{n+1}}{\sqrt{n}+2}$ с $\frac{1}{n^{1/6}}$, понеже общият член е знакопроменлив. Затова:

$\sum (-1)^n \frac{\sqrt[3]{n+1}}{\sqrt{n}+2} = \sum \frac{(-1)^n}{n^{1/6}} + ?$ За да определим ? – например:

$$\frac{\sqrt[3]{n+1}}{\sqrt{n}+2} = \frac{\sqrt[3]{n}\left(1+\frac{1}{n}\right)^{1/3}}{\sqrt{n}\left(1+\frac{2}{\sqrt{n}}\right)} = \frac{1}{n^{1/6}}\left(1+\frac{1}{n}\right)^{1/3}\left(1+\frac{2}{n}\right)^{-1} = \frac{1}{n^{1/6}}\left(1+O\left(\frac{1}{n}\right)\right)\left(1+O\left(\frac{1}{n}\right)\right) =$$

$= \frac{1}{n^{1/6}}\left(1+O\left(\frac{1}{n}\right)\right) = \frac{1}{n^{1/6}}+O\left(\frac{1}{n^{7/6}}\right)$. Тогава

$\sum (-1)^n \frac{\sqrt[3]{n+1}}{\sqrt{n}+2} = \sum\left(\frac{(-1)^n}{n^{1/6}}+O\left(\frac{1}{n^{7/6}}\right)\right)$ и $\sum \frac{(-1)^n}{n^{1/6}}$ – сх., $\sum O\left(\frac{1}{n^{7/6}}\right)$ – абс. сх.

Окончателно: $\sum$ е сходящ.

㉑ $$\sum \cos\left(\pi n+\frac{\pi}{6}\right)\ln\left(1+\frac{2}{n}\right) = \sum (-1)^n\cos\frac{\pi}{6}\ln\left(1+\frac{2}{n}\right) = \frac{\sqrt{2}}{2}\sum (-1)^n\ln\left(1+\frac{2}{n}\right)$$

$$= \frac{\sqrt{2}}{2}\sum (-1)^n\left(\frac{2}{n}+O\left(\frac{1}{n^2}\right)\right) = \sqrt{2}\sum_1 \frac{(-1)^n}{n} + \frac{\sqrt{2}}{2}\sum_2 (-1)^n O(1/n^2)$$

$\sum_1$ – сходящ, следва от критерия на Лайбниц

$\sum_2$ – абс. сх., понеже $\sum 1/n^2 \rightarrow$ сх. Окончателно $\sum$ – сх.

㉒ $\sum \frac{(-1)^n}{n^\alpha-(-1)^n}$

Разглеждаме два случая:

① $\alpha>0$. Тогава $n^\alpha-(-1)^n \sim n^\alpha$ при $n\to\infty$, но не можем да заместим $n^\alpha-(-1)^n$ с n^α, понеже редът е знакопроменлив.

Но знаем, че $\sum \frac{(-1)^n}{n^\alpha}$ е сходящ ($\Leftarrow$ от критерия на Лайбниц)

Затова отделяме този ред, т.е. $\sum \frac{(-1)^n}{n^\alpha-(-1)^n} = \sum \frac{(-1)^n}{n^\alpha} + ?$

За определянето на ? например:

$$\sum \frac{(-1)^n}{n^\alpha-(-1)^n} = \sum \frac{(-1)^n}{n^\alpha}\cdot\frac{1}{1-\frac{(-1)^n}{n^\alpha}} = \sum \frac{(-1)^n}{n^\alpha}\left(1+\frac{(-1)^n}{n^\alpha}+o\left(\frac{1}{n^\alpha}\right)\right) =$$

$$= \underbrace{\sum \frac{(-1)^n}{n^\alpha}}_{\text{сх.}} + \underbrace{\sum\left(\frac{(-1)^{2n}}{n^{2\alpha}}+o\left(\frac{1}{n^{2\alpha}}\right)\right)}_{= \sum\left(\frac{1}{n^{2\alpha}}+o\left(\frac{1}{n^{2\alpha}}\right)\right)} \sim \sum \frac{1}{n^{2\alpha}}$$ понеже

$\frac{1}{n^{2\alpha}}+o\left(\frac{1}{n^{2\alpha}}\right) \sim \frac{1}{n^{2\alpha}}$ и $\frac{1}{n^{2\alpha}}+o\left(\frac{1}{n^{2\alpha}}\right) > 0$, т.е. можем да заменим $\frac{1}{n^{2\alpha}}+o\left(\frac{1}{n^{2\alpha}}\right)$ с $\frac{1}{n^{2\alpha}}$.

Сега $\sum \frac{1}{n^{2\alpha}}$ е сх. при $2\alpha>1$, т.е. $\alpha>1/2$ и разходящ при $\alpha \leq 1/2$, т.е. при $0<\alpha\leq 1/2$.

② $\alpha<0$

Тогава $\frac{(-1)^n}{n^\alpha-(-1)^n} \xrightarrow[n\to\infty]{} -1$, понеже $n^\alpha \xrightarrow[n\to\infty]{} 0$, т.е. $a_n \not\rightarrow 0$

$\Rightarrow$ редът е разходящ.

Окончателно $\sum$ е сх. при $\alpha>1/2$ и разх. при $\alpha \leq 1/2$.

㉓ $$\sum \ln\left(1+\frac{(-1)^{n+1}}{\sqrt{n}}\right) = \sum\left[\frac{(-1)^{n+1}}{\sqrt{n}}-\frac{1}{2}\frac{(-1)^{2(n+1)}}{n}+o(1/n)\right]$$

Тук използваме развитието $\ln(1+x) = x-\frac{x^2}{2}+o(x^2)$.

$\sum \frac{(-1)^{n+1}}{\sqrt{n}}$ е сх. ($\Leftarrow$ Лайбниц)

$-\frac{1}{2}\sum\left(\frac{1}{n}+o\left(\frac{1}{n}\right)\right) \sim \sum \frac{1}{n}$, понеже $\frac{1}{n}+o\left(\frac{1}{n}\right) > 0$ и $\frac{1}{n}+o\left(\frac{1}{n}\right) \sim 1/n$

и понеже $\sum \frac{1}{n}$ е разх. $\Rightarrow \sum \ln\left(1+\frac{(-1)^{n+1}}{\sqrt{n}}\right)$ е разх.

㉔ $$\sum\left[\frac{\sin n}{\sqrt[3]{n^2}}-\sin\left(\frac{\sin n}{\sqrt[3]{n^2}}\right)\right] = \sum_1 \frac{\sin n}{n^{2/3}} - \sum_2 \sin\left(\frac{\sin n}{n^{2/3}}\right)$$

$\sum_1$ е сходящ ($\Leftarrow$ от критерия на Дирихле, тъй като $\left|\sum_1^N \sin n\right| < M$ и $\frac{1}{n^{2/3}} \searrow 0$.

$$\sum_2 = \sum\left(\frac{\sin n}{n^{2/3}}+O\left(\frac{1}{n^2}\right)\right)$$

Тук използваме $\sin x = x-\frac{x^3}{6}+o(x^3) = x+O(x^3)$

$\sum \frac{\sin n}{n^{2/3}} \rightarrow$ сх., $\sum O(1/n^2) \rightarrow$ абс. сх.

Окончателно $\sum$ е сходящ.

㉕ $\sum (-1)^n\left(\operatorname{arctg}\frac{1}{n}-\arcsin\frac{1}{n}\right)$

$\operatorname{arctg} x = x+O(x^3) \Rightarrow \operatorname{arctg} 1/n = \frac{1}{n}+O(1/n^3)$

За развитието на $\arcsin x \Rightarrow$ диференциране, т.е.

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-\frac{1}{2}} = 1+\frac{1}{2}x^2+\dots \Rightarrow$$

$$\arcsin x = \int\left(1+\frac{1}{2}x^2+\dots\right)dx = x+\frac{x^3}{6}+\dots = x+O(x^3)$$

$$\Rightarrow \arcsin\frac{1}{n} = \frac{1}{n}+O\left(\frac{1}{n^3}\right) \Rightarrow$$

$$\operatorname{arctg}\frac{1}{n}-\arcsin\frac{1}{n} = O\left(\frac{1}{n^3}\right) \Rightarrow$$

$$\sum (-1)^n\left(\operatorname{arctg}\frac{1}{n}-\arcsin\frac{1}{n}\right) = \underbrace{\sum (-1)^n O(1/n^3)}_{\text{абс. сх.}}$$

Окончателно $\sum$ е абс. сх. $\Rightarrow$ сходящ.